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研究兴趣

公平分配、算法博弈论、机制设计、在线算法、近似算法。

教育背景

2021 – 2025 **博士，计算机科学**，澳门大学，澳门特别行政区
博士论文：*Fairly Allocating Indivisible Chores to Symmetric and Asymmetric Agents*
导师：吴晓伟教授

2019 – 2020 **硕士，城市信息学**，伦敦国王学院，英国伦敦
论文导师：Angus Roberts 博士

2014 – 2019 **学士，城乡规划**，武汉大学，中国武汉
辅修：地理信息科学

研究经历

2025 – 至今 **博士后研究员**，南洋理工大学，新加坡
导师：贝小辉教授

2021 **研究助理**，澳门大学，澳门特别行政区
导师：吴晓伟教授

荣誉与奖学金

2025 **最佳学生论文奖**，*Conference on Web and Internet Economics (WINE)*
225 篇投稿中唯一获奖论文

2025 **太湖奖学金**，澳门大学
粤港澳大湾区太湖奖学金

2021 – 2025 **澳门大学博士助学金**，澳门大学

2019 **优秀学士学位论文奖**，武汉大学
本科毕业论文获评城市设计学院前 5%

工作论文 (α-β)：作者按字母序 ✉：通讯作者

[W4] **Degree-bounded Online Bipartite Matching Revisited: OCS vs. Ranking**
(α-β) Yilong Feng, Haolong Li, Xiaowei Wu, Shengwei Zhou.
投稿至 *Mathematics of Operations Research*

[W3] **Efficient EFX Allocations for Chores**
(α-β) Zehan Lin, Xiaowei Wu, Shengwei Zhou.
Artificial Intelligence 返修

[W2] **An FPTAS for 7/9-Approximation to Maximin Share Allocations**
(α-β) Xin Huang, Shengwei Zhou.
工作论文

[W1] **When Maximum Nash Welfare Becomes Strongly Fair: Bi-valued Goods**
(α-β) Zehan Lin, Xiaowei Wu, Shengwei Zhou.
工作论文

会议论文 (α-β)：作者按字母序

[C17] **Fair Division by Contribution: A Shapley Value Perspective**
(α-β) Xiaohui Bei, Pinyan Lu, Xiaowei Wu, Shengwei Zhou.
Proceedings of the 27th ACM Conference on Economics and Computation (EC), 2026, Forthcoming.

[C16] **Allocating Chores with Restricted Additive Costs: Achieving EFX, MMS, and Efficiency Simultaneously**
(α-β) Zehan Lin, Xiaowei Wu, Shengwei Zhou.
Proceedings of the ACM on Web Conference (WWW), pages 146-156, Apr 2026. doi:10.1145/3774904.3792300

[C15] **All-but-One MMS Allocation for Chores**
(α-β) Jiawei Qiu, Xiaowei Wu, Cong Zhang, Shengwei Zhou.
Proceedings of the ACM on Web Conference (WWW), pages 157-168, Apr 2026. doi:10.1145/3774904.3792305

[C14] **Degree-bounded Online Bipartite Matching Revisited: OCS vs. Ranking**
(α-β) Yilong Feng, Haolong Li, Xiaowei Wu, Shengwei Zhou.
Proceedings of the 21st Conference on Web and Internet Economics (WINE), pages 747–764, Dec 2025. doi:10.1007/978-3-032-18660-7_39 ★ 最佳学生论文

[C13] **Incentive Analysis of Collusion in Fair Division**
(α-β) Haoqiang Huang, Biaoshuai Tao, Mingwei Yang, Shengwei Zhou.
Proceedings of the 21st Conference on Web and Internet Economics (WINE), pages 392–410, Dec 2025. doi:10.1007/978-3-032-18660-7_21

[C12] **When is Truthfully Allocating Chores no Harder than Goods?**
(α-β) Bo Li, Biaoshuai Tao, Fangxiao Wang, Xiaowei Wu, Mingwei Yang, Shengwei Zhou.
Proceedings of the 18th International Symposium on Algorithmic Game Theory (SAGT), pages 247-264, Sep 2025. doi:10.1007/978-3-032-03639-1_14.

[C11] **Approximately EFX and PO Allocations for Bivalued Chores**
(α-β) Zehan Lin, Xiaowei Wu, Shengwei Zhou.
Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI), pages 3952-3960, Aug 2025. doi:10.24963/ijcai.2025/440.

[C10] **A Little Subsidy Ensures MMS Allocation for Three Agents**
(α-β) Xiaowei Wu, Quan Xue, Shengwei Zhou.
Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI), pages 4073-4081, Aug 2025. doi:10.24963/ijcai.2025/454.

[C9] **Revisiting Proportional Allocation with Subsidy: Simplification and Improvements**
(α-β) Xiaowei Wu, Quan Xue, Shengwei Zhou.
Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI), pages 4082-4090, Aug 2025. doi:10.24963/ijcai.2025/455.

[C8] **Pure Nash Equilibrium of Weight Picking Sequence Protocol is WEF1 for Two Agents**
(α-β) Rufan Bai, Huahua Miao, Xiaowei Wu, Cong Zhang, Shengwei Zhou.
Proceedings of the 19th International Joint Conference, IJTCS-FAW, pages 238–251, Jun 2025. doi:10.1007/978-981-96-8312-3_18.

[C7] **Tree Splitting Based Rounding Scheme for Weighted Proportional Allocations with Subsidy**
(α-β) Xiaowei Wu, Shengwei Zhou.
Proceedings of the 20th Conference on Web and Internet Economics (WINE), pages 295–313, Dec 2024. doi:10.1007/978-3-032-08560-3_17.

[C6] **On the Existence of EFX (and Pareto-Optimal) Allocations for Binary Chores**
(α-β) Biaoshuai Tao, Xiaowei Wu, Ziqi Yu, Shengwei Zhou.
Proceedings of the 18th International Joint Conference, IJTCS-FAW, pages 33-52, Aug 2024. doi:10.1007/978-981-97-7752-5_3.

[C5] **One Quarter Each (on Average) Ensures Proportionality**
(α-β) Xiaowei Wu, Cong Zhang, Shengwei Zhou.
Proceedings of the 20th Conference on Web and Internet Economics (WINE), pages 582-599, Dec 2023. doi:10.1007/978-3-031-48974-7_33.

[C4] **Improved Competitive Ratio for Edge-Weighted Online Stochastic Matching**
Guoliang Qiu, Yilong Feng, Shengwei Zhou, Xiaowei Wu.
Proceedings of the 20th Conference on Web and Internet Economics (WINE), pages 527-544, Dec 2023. doi:10.1007/978-3-031-48974-7_30.

[C3] **Multi-agent Online Scheduling: MMS Allocations for Indivisible Items**
Shengwei Zhou, Rufan Bai, Xiaowei Wu.
Proceedings of the 40th International Conference on Machine Learning (ICML), pages 42506-42516, Jul 2023. doi:10.5555/3618408.3620197.

[C2] **Weighted EF1 Allocations for Indivisible Chores**
(α-β) Xiaowei Wu, Cong Zhang, Shengwei Zhou.
Proceedings of the 24th ACM Conference on Economics and Computation (EC), pages 1155, Jul 2023. doi:10.1145/3580507.3597763.

[C1] **Approximately EFX Allocations for Indivisible Chores**
Shengwei Zhou, Xiaowei Wu.
Proceedings of the 31st International Joint Conference on Artificial Intelligence (IJCAI), pages 783-789, Aug 2022. doi:10.24963/ijcai.2022/110.

期刊论文 (α-β)：作者按字母序

[J4] **Subsidised Proportionality in Fair Allocation**
(α-β) Xiaowei Wu, Cong Zhang, Shengwei Zhou✉.
Operations Research, Forthcoming.
会议版本：WINE 23（C5）、WINE 24（C7）和 IJCAI 25（C9）。

[J3] **Weighted EF1 Allocations for Indivisible Chores**
(α-β) Xiaowei Wu, Cong Zhang, Shengwei Zhou.
Artificial Intelligence (AIJ), 347:104386, Oct 2025. doi:10.1016/j.artint.2025.104386.
会议版本：EC 23（C2）。

[J2] **On the Existence of EFX (and Pareto-Optimal) Allocations for Binary Chores**
(α-β) Biaoshuai Tao, Xiaowei Wu, Ziqi Yu, Shengwei Zhou✉.
Theoretical Computer Science (TCS), 1042:115248, Jul 2025.
doi:10.1016/j.tcs.2025.115248. 会议版本：IJTCS-FAW 24（C6）。

[J1] **Approximately EFX Allocations for Indivisible Chores**
Shengwei Zhou, Xiaowei Wu.
Artificial Intelligence (AIJ), 326:104037, Jan 2024. doi:10.1016/j.artint.2023.104037.
会议版本：IJCAI 22（C1）。

教学经历

2022 秋 – 2024 **助教**，*University of Macau*
CISC7007 Design and Analysis of Algorithm（研究生课程）

2022 春 – 2025 **助教**，*University of Macau*
CISC2006 Algorithm Design and Analysis（本科课程）

2021 秋 **助教**，*University of Macau*
CISC3027 Special Topics in Computer and Information Science（本科课程）

学术报告

2026 年 7 月	Fair Division by Contribution: A Shapley Value Perspective *ACM EC, Rome*
2025 年 8 月	Revisiting Proportional Allocation with Subsidy: Simplification and Improvements *IJCAI, Guangzhou*
2025 年 4 月	Revisiting Proportional Allocation with Subsidy: Simplification and Improvements *IOTSC Postgraduate Forum, University of Macau*
2024 年 10 月	Fair Allocations of Chores with Subsidy *Invited Talk, Kyushu University*
2024 年 8 月	On the Existence of EFX (and Pareto-Optimal) Allocations for Binary Chores *IJTCS-FAW, Hong Kong*
2024 年 3 月	PROP Allocation for Indivisible Chores with Subsidy *Invited Talk, Shanghai Jiao Tong University*
2023 年 12 月	Weighted PROP Allocations with Subsidy *Workshop on Theoretical Computer Science, University of Macau*
2023 年 12 月	Recent Development of Weighted Fair Allocation *Invited Talk, Wuhan University*
2023 年 8 月	Weighted EF1 Allocations for Indivisible Chores *Young PhD Forum, IJTCS-FAW, Macao*
2023 年 7 月	Weighted EF1 Allocations for Indivisible Chores *ACM EC, London*
2022 年 11 月	Approximately EFX Allocations for Indivisible Chores *IJCAI-China, Shenzhen*

学术服务

程序委员会委员	AAMAS (2026), EC (2026), IJCAI (2026), WINE (2026), WINE (2025)
会议（子）审稿人	NeurIPS (2026), SODA (2026), SOSA (2026), TAMC (2026), WWW (2025), ISAAC (2024), SODA (2023)
期刊审稿人	Information Sciences, Journal of Autonomous Agents and Multi-Agent Systems (JAAMAS), SIAM Journal on Discrete Mathematics (SIDMA), Theoretical Computer Science (TCS)
组织委员会	IJTCS-FAW (2023)